Agathe **Fernandes Machado**

PhD student · Department of Mathematics
Université du Québec à Montréal (UQAM), Montreal
201 Av. du Président-Kennedy -- QC H2X 3Y7 Montreal -- Canada

she/her | fernandes_machado.agathe@courrier.uqam.ca | fer-agathe.github.io | fer-agathe

Research topics

- Machine Learning Models,
- Trustworthy Artificial Intelligence (AI)
  - Algorithmic Fairness via Causal Inference and Optimal Transport Theory,
  - Explainability (XAI): Cooperative Game Theory, Counterfactual Explanations,
  - Model Reliability: Calibration,
  - Uncertainty Quantification: Conformal Prediction,
- Applied Mathematics: Actuarial Science (Climate Risks).

Technical Skills

- Programming Languages: Python, R, MATLAB, SQL, Java,
- Machine Learning & Data Science: scikit-learn, PyTorch, Pandas, NumPy,
- Tools & Platforms: GitHub, Jupyter, VS Code, Linux.

Education

PhD student in Mathematics — *Montréal (Canada)*
Université du Québec à Montréal — *2023 – 2027*
- Thesis title: Algorithmic Fairness and Discrimination in Predictive Models;
- Supervisors: Arthur Charpentier, Ewen Gallic;
- **PhD Courses** at Mila Institute (Montreal)
  - Probabilistic and Graphical Models (Simon Lacoste-Julien, Fall 2023): Directed and Undirected Graphical Models, Gaussian Networks, Approximate Inference,
  - Representation Learning (Aaron Courville, Winter 2024): CNNs, RNNs, LSTM, Transformers, GANs, Applications on PyTorch.

Master's in Actuarial Science (with High Honors) — *Brest (France)*
Euria, Université de Bretagne Occidentale — *2021 – 2023*
Double Degree with IMT Atlantique

Master's from a Generalist Engineering School — *Brest (France)*
IMT Atlantique — *2019 – 2023*
Major: Mathematical and Computational Engineering: Statistical Learning, Stochastic Processes and Numerical Optimization

Classes Préparatoires aux Grandes Ecoles, Physics, Chemistry, and Engineering Sciences (Equivalent to a Bachelor's degree) (with Honors) — *Rennes (France)*
Lycée Chateaubriand — *2017 – 2019*
- Major: Mathematics, Physics, Chemistry;
- Second year in PC*, called "star class".

Work experience

Internship in actuarial science (1 year) - Reinsurance pricing and Climate modeling — *Nantes (France)*
Generali, Insurance company — *2022 – 2023*
- Programming: RMS (natural disaster modeling), Python, R, PySpark.

Internship in actuarial science (3 months) - Climate modeling (Drought risk) — *Paris (France)*
Seabird, Consulting firm in insurance/finance — *2023*
- Extreme value theory: analysis of clay shrinkage and swelling risk based on drought index values (KBDI, SSWI) and portfolio claims experience;
- Participation in a mission to merge loan insurance products.

Internship *Data scientist (4 months)* — *Vannes (France)*

CRÉDIT AGRICOLE, BANK, MARKETING RESEARCH AND DEVELOPMENT DEPARTMENT — *2021*

- Optimization of scores from machine learning methods based on tracking and customer profile data (XGBoost, Random Forest, etc.);
- Programming: SAS Guide, Python and Big Data tools.

Internship JavaScript Development (1 month) — *Vannes (France)*

DAWIZZ, START-UP IN IT — *2019*

Research projects

Visiting researcher (2 months) - Interpretability/XAI — *Sydney (Australia)*

UNSW BUSINESS SCHOOL - SCHOOL OF RISK AND ACTUARIAL STUDIES — *2026*

- Invited by Associate Professor Fei Huang.
- Application of interpretability methods to insurance pricing models.

Visiting researcher (4 months) - Causal Inference: Potential Outcomes Framework — *Marseille (France)*

AIX-MARSEILLE SCHOOL OF ECONOMICS (AMSE) — *2025/2026*

- Invited by Assistant Professor Ewen Gallic, in collaboration with Professor Emmanuel Flachaire.
- Robustness of Average Treatment Effect (ATE) estimators to the miscalibration of propensity scores (with Inverse Probability Weighting and Double Machine Learning).

Visiting researcher (2 months) - Discrimination in Mortality Scores — *Paris (France)*

MILLIMAN FRANCE - CONSERVATOIRE NATIONAL DES ARTS ET MÉTIERS (CNAM) — *2025*

- Theoretical development and implementation of a mitigation technique for a group-fairness criterion, Predictive Parity;
- Writing an article intended for submission within the year 2026.

Development of the Python package `EquiPy` — *Montréal (Canada)*

UNIVERSITÉ DU QUÉBEC À MONTRÉAL — *2023*

- Post-processing method to mitigate discrimination in the predictions of a machine learning model, Sequential Fairness;
- Documentation: `https://equilibration.github.io/equipy/`.

Actuarial research thesis (1 year) - Reinsurance pricing and Climate modeling — *Nantes (France)*

GENERALI, EURIA (UNIVERSITÉ DE BRETAGNE OCCIDENTALE) — *2022 – 2023*

- Thesis title: Marginal contribution of industrial sites to the reinsurance cost of an excess of loss per event treaty;
- Application of reinsurance pricing methods for an excess of loss per event treaty to industrial risks (companies) using Monte-Carlo methods.

Actuarial research project with Sia Partners (1 year) - Climate modeling — *Paris (France)*

SIA PARTNERS, EURIA (UNIVERSITÉ DE BRETAGNE OCCIDENTALE) — *2021 – 2022*

- Projection of drought risk in France, measured by the KBDI index, using temperature and precipitation data (`https://data.nasa.gov`) and IPCC scenarios;
- Implementation of a climate scenario generator with R.

Publications

Published articles

1. Fernandes Machado, A., Charpentier, A., & Gallic, E. (2025). Optimal Transport on Categorical Data for Counterfactuals using Compositional Data and Dirichlet Transport. *34th International Joint Conference on Artificial Intelligence* (IJCAI 2025).

2. Il Idrissi, M., Fernandes Machado, A., & Charpentier, A. (2025). Beyond Shapley Values: Cooperative Games for the Interpretation of Machine Learning Models. *34th International Joint Conference on Artificial Intelligence* (IJCAI 2025) *Workshop on Explainable Artificial Intelligence (XAI)*.

3. Fernandes Machado, A., Charpentier, A., & Gallic, E. (2025). Sequential conditional transport on probabilistic graphs for interpretable counterfactual fairness. *The 39th Annual AAAI Conference on Artificial Intelligence* (AAAI 2025).

4. Fernandes Machado, A., Charpentier, A., Flachaire, E., Gallic, E. & Hu, F. (2024). Post-Calibration Techniques: Balancing Calibration and Score Distribution Alignment. *Thirty-Eighth Annual Conference on Neural Information Processing Systems* (NeurIPS 2024) *Bayesian Decision Uncertainty (BDU) Workshop*.

Preprints

1. Il Idrissi, M., Fernandes Machado, A., Gallic, E., & Charpentier, A. (2025). *Unveil Sources of Uncertainty: Feature Contribution to Conformal Prediction Intervals*. https://arxiv.org/abs/2505.13118
2. Fernandes Machado, A., Grondin, S., Ratz, P., Charpentier, A., & Hu, F. (2025). *EquiPy: Sequential Fairness using Optimal Transport in Python*. https://arxiv.org/abs/2503.09866
3. Fernandes Machado, A., Charpentier, A., Flachaire, E., Gallic, E., & Hu, F. (2024). *Probabilistic scores of classifiers, calibration is not enough*. https://arxiv.org/abs/2408.03421
4. Fernandes Machado, A., Charpentier, A., Flachaire, E., Gallic, E., & Hu, F. (2024). *From uncertainty to precision: Enhancing binary classifier performance through calibration*. https://arxiv.org/abs/2402.07790
5. Fernandes Machado, A., Hu, F., Ratz, P., Gallic, E., & Charpentier, A. (2024). *Geospatial disparities: A case study on real estate prices in paris*. https://arxiv.org/abs/2401.16197

Conferences, Workgroups, Seminars

1. 03/2026 Workshop When Uncertainty Quantification meets Causal Modelling and Inference (SCAI, Paris): *Causal Mediation Analysis via Sequential Transport to Assess Counterfactual Fairness.*
2. 03/2026 Workshop AI in risk assessment and mitigation (ICMS & Bayes centre, Edinburgh): *Measuring Fairness through Calibration in Insurance Scoring Algorithms.*, https://github.com/fer-agathe/calibration_fairness_insurance.git.
3. 02/2026 APSPP/SINCLAIR Seminar (EDF Lab Chatou, Paris): *Unveil Sources of Uncertainty in Machine Learning Models: Feature Contribution to Conformal Prediction Intervals.*
4. 02/2026 Causal Decision Making, Algorithms and Fairness Workshop (Rotterdam School of Management, Erasmus University): *Assessing Counterfactual Fairness via (Marginally) Optimal Transport.*
5. 02/2026 Econometrics and Big Data Seminar (Aix-Marseille School of Economics, France): *Assessing Counterfactual Fairness via (Marginally) Optimal Transport.*
6. 01/2026 Mathematics Seminar (University of Bretagne Occidentale, Brest, France): *Unveil Sources of Uncertainty in Machine Learning Models: Feature Contribution to Conformal Prediction Intervals.*
7. 01/2026 UvA, SIAS Seminar (University of Amsterdam): *Assessing Counterfactual Fairness via (Marginally) Optimal Transport.*
8. 01/2026 QARMA Seminar (University of Aix-Marseille, France): *Assessing Counterfactual Fairness via (Marginally) Optimal Transport.*
9. 10/2025 Quantact Seminar on Actuarial and Financial Mathematics (ULaval, Quebec): *Unveil Sources of Uncertainty in Machine Learning Models: Feature Contribution to Conformal Prediction Intervals.*
10. 10/2025 CIRRELT Monthly Reading group at Polytechnique Montreal: *Assessing Counterfactual Fairness via Optimal Transport Theory.*
11. 09/2025 PrivSec Days 2025 on Privacy and Security of AI (Montreal): *Assessing Counterfactual Fairness via Optimal Transport Theory.*
12. 08/2025 34th International Joint Conference on Artificial Intelligence IJCAI 2025 (Montreal): *Optimal Transport on Categorical Data for Counterfactuals using Compositional Data and Dirichlet Transport.*, Oral Presentation.
13. 08/2025 Doctoral Consortium - 34th International Joint Conference on Artificial Intelligence IJCAI 2025 (Montreal): *Ensuring Reliable and Transparent Algorithmic Fairness through Optimal Transport and Uncertainty Quantification.*, Oral Presentation, https://2025.ijcai.org/doctoral-consortium-program/.
14. 08/2025 Actuarial and statistical summer seminar (Université du Québec à Montréal): *EquiPy: A Python Package implementing Sequential Fairness with Optimal Transport.*, https://github.com/TommyMastro/Seminaire_actu_stats_UQAM.git.
15. 06/2025 Workshop Calibrating prediction uncertainty, statistics and machine learning perspectives (Isaac Newton Institute, Cambridge): *Beyond Calibration of Probabilistic Classifier Outputs.*, https://www.newton.ac.uk/event/rclw02/.
16. 05/2025 STATQAM Research Day (Université du Québec à Montréal, Montréal): *Assessing Counterfactual Fairness via (Marginally) Optimal Transport*, https://github.com/fer-agathe/statqam_research_day.git.

17. 02/2025 Actuarial and Financial Mathematics Conference (AG Campus, Brussels): *Predicting Unobserved Multi-Class Sensitive Attributes: Enhancing Calibration with Nested Dichotomies for Fairness.*, https://github.com/fer-agathe/AFM_2025.git.
18. 01/2025 Workshop (Milliman France, Paris): *Analyzing Discrimination in Mortality Scores.*
19. 01/2025 Seminar PSPP (EDF Lab Chatou, Paris): *Challenging the performance of binary classifiers through the notion of calibration.*
20. 12/2024 NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty (Vancouver): *Post-Calibration Techniques: Balancing Calibration and Score Distribution Alignment.*, https://hal.science/hal-04916151/.
21. 08/2024 Actuarial and statistical summer seminar (Université du Québec à Montréal): *Probabilistic scores of classifiers, calibration is not enough.*, https://github.com/TommyMastro/Seminaire_actu_stats_UQAM.git.
22. 08/2024 Seminar *Seminario de Matemáticas Aplicadas* (Quantil, Colombia, remote): package `EquiPy`, https://github.com/fer-agathe/quantil_seminar.git.
23. 06/2024 Insurance Data Science Conference (Stockholm University): *Probabilistic scores of classifiers, calibration is not enough.*, https://github.com/fer-agathe/IDSC_2024.git.
24. 05/2024 Annual Conference *Société Canadienne de Science Economique* 2024 (HEC Montréal): *From uncertainty to precision: Enhancing binary classifier performance through calibration.*, https://github.com/fer-agathe/scse_2024.git.
25. 05/2024 Workshop on Fairness and Discrimination in Insurance 2024 (Université Laval, Québec): package `EquiPy`.
26. 04/2024 Science Research Day 2024 (Université du Québec à Montréal): 4-minute presentation of the research project, https://github.com/fer-agathe/projet_recherche_court.git.
27. 05/2024 Workshop in Insurance Mathematics 2024 (Concordia University, Montréal): poster presentation, package `EquiPy`, https://github.com/fer-agathe/WIM_2024_equipy.git.

Scholarships

PBEEE Excellence Scholarships for International Students (declined) — *Fonds De Recherche du Québec*
PROGRAM: PHD – FIELD: NATURE AND TECHNOLOGY – DISCIPLINE: STATISTICAL LEARNING — *2026/2027*

Scholarship for a Research internship in France (CRM) — *CRM, Montreal*
RESEARCH INTERNSHIP SCHOLARSHIPS IN FRANCE FOR CENTRE DE RECHERCHES MATHÉMATIQUES (CRM) STUDENTS — *2025*

Mobility scolarship (UQAM) — *Université du Québec à Montréal*
THE INTERNATIONAL MOBILITY AND SHORT STAYS OUTSIDE QUEBEC PROGRAM OF THE MINISTRY OF HIGHER EDUCATION IS INTENDED FOR UQAM STUDENTS WHO WISH TO UNDERTAKE A STUDY STAY OF AT LEAST THREE WEEKS OUTSIDE CANADA. — *2025*

PhD scolarship Renewal (OBVIA) — *Université du Québec à Montréal*
SUPPORTING THE NEXT GENERATION SCHOLARSHIP PROGRAM 2025, INTERNATIONAL OBSERVATORY ON THE SOCIETAL IMPACTS OF AI AND DIGITAL TECHNOLOGIES — *2025/2026*

Scholarship for a Research internship in France (CRM) — *CRM, Montreal*
RESEARCH INTERNSHIP SCHOLARSHIPS IN FRANCE FOR CENTRE DE RECHERCHES MATHÉMATIQUES (CRM) STUDENTS — *2024*

PhD scolarship (OBVIA) — *Université du Québec à Montréal*
SUPPORTING THE NEXT GENERATION SCHOLARSHIP PROGRAM 2024, INTERNATIONAL OBSERVATORY ON THE SOCIETAL IMPACTS OF AI AND DIGITAL TECHNOLOGIES — *2024/2025*

Teaching experience

Statistics

Interpretability Methods for Machine Learning models (6 hours of lectures) *EURIA, UBO*
MASTER'S DEGREE YEAR 1 (40 STUDENTS) *2026*
- Introduction to Git, GitHub and Visual Studio Code;
- Overview of interpretability methods for machine learning models and application of SHAP and Partial Dependence Plots (PDP) in Python.

Statistical learning (3 hours of lectures and 3 hours of laboratory sessions) *Université du Québec à Montréal*
FIRST CYCLE (17 STUDENTS) *2024*
- Linear models, polynomial regression, linear classification (logistic and multinomial regressions), variable selection methods (best subset method, forward, backward, and stepwise), regression regularization methods (Lasso and Ridge), applications using R;
- Website: https://etudier.uqam.ca/STT3030.

IT

Introduction to Python (8 hours of laboratory sessions) *IMT Atlantique*
SECONDARY STUDENTS *2021*
As part of an academic project titled "Sustainable Development and Social Engagement," along with 5 other students, we taught 4 Python classes to Secondary 1 to 3 students, targeting girls to encourage gender equality in technological and IT professions.

Student supervision

Bachelor

Allison Lara Nieva *Université du Québec à Montréal*
RESEARCH INTERNSHIP OF 3 MONTHS *2025*
Combining decision trees and logistic regression models;
Co-supervision of internship with Ewen Gallic and Arthur Charpentier.

Iryna Voitsitska *Université du Québec à Montréal*
RESEARCH INTERNSHIP OF 3 MONTHS *2025*
Using Optimal Transport theory to estimate counterfactuals within causal inference;
Co-supervision of internship with Ewen Gallic and Arthur Charpentier.

Ana-Maria Patrón Piñerez *Université du Québec à Montréal*
RESEARCH INTERNSHIP OF 3 MONTHS *2024*
Algorithmic Fairness: Bayesian methods to predict ethnicity following Colorado legislation SB21-169;
Co-supervision of internship with Arthur Charpentier.

Participating in collective tasks

Peer Review Activities

1. 2026 Reviewer for the International Conference on Machine Learning (ICML) 2026.
2. 2025/2026 Reviewer for the AISTATS 2026 conference.
3. 2024/2025 Reviewer for the European Actuarial Journal (EAJ).

Research

1. 05/2025 Organization of the 2025 Scientific Day of the OBVIA Community (HEC, Montréal): organization and development of a creative method to present and connect the research work of student members of the OBVIA institute.
2. 09/2024 Organization of the Quantact seminar (Université du Québec à Montréal): Presentation by Adel Cherchali (Milliman, France) on applications of Large Language Models in insurance.
3. 05/2024 Co-organization of Quantact Summer Day (Université de Montréal): A day of presentations by students enrolled in master's or PhD programs in actuarial sciences and financial mathematics, from universities across Quebec (Université de Montréal, Université du Québec à Montréal, HEC Montréal, Université de Sherbrooke, Université Laval and Concordia University).